Decision Models for Binary Financial Document Triage: Adaptation, Missed Documents, and Processing Cost

Lucca Pereira Martins Matthew Ball

Abstract

Financial document triage trades reduced analyst reading against missed information and processing cost. We compare seven decision-model, finance-encoder and lexical systems using binary language-model reference labels. The SEC-dominated corpus contains 25,000 training, 1,000 selection, 500 calibration and 2,000 test documents. Core trained systems share 4,000 documents. Laya LoRA achieves macro F1 0.7843 versus 0.7645 for word TF-IDF/logistic; the paired 95% difference interval $[-0.0022, 0.0423]$ includes zero. They miss 268 and 188 of 1543 review-worthy references while skipping 32.4% and 24.6% of documents. Laya's warm mean inference latency is 64.9 times the lexical classifier's. Training-size curves reach 25,000 documents; Laya's 16k-to-25k F1 increment has an interval including zero. Calibration-selected policies quantify the workload–miss tradeoff. Results measure agreement with Luna labels on the shared test cohort, not analyst-validated materiality or performance on a fresh sample. We release predictions and independent CPU replay.

1 Introduction

Financial document triage routes incoming information to automatic skipping or analyst review. Its value depends on how much reading it removes, how much important information it misses, and its processing cost.

We classify public financial documents as *routine* or *review-worthy* using their content alone. Potentially significant but unresolved developments go to review; uncertainty is metadata, not a third destination. Labels supplied by a language-model teacher define this review convention. They do not establish investment materiality, which can depend on company history and portfolio context.

We ask whether adapting a pretrained decision model improves triage relative to a finance encoder and a lexical classifier, and whether any gain warrants its processing cost. Seven systems form the core comparison: native Laya, Laya with head adaptation or low-rank adaptation (LoRA), frozen FinBERT with logistic regression, FinBERT with LoRA, word TF-IDF with logistic regression, and a majority baseline. Trainable core systems share 4,000 documents; larger nested subsets measure sample efficiency.

The contribution is a financial-domain comparison with event-group controls, matched pre-update models, and reproducible error, workload and resource measurements. The binary study follows a three-class exploration and reuses its test documents. We froze binary choices before reevaluation, but prior inspection limits the independence of the resulting evidence.

2 Related work

Financial relevance filtering is established. Ahbali et al. [7] use a TF-IDF/SVM relevance filter before corporate credit-event analysis. Del Corro and Hoffart [8] derive financial-news relevance from stock-prediction attention. FinLin [9] provides multi-source, expert-annotated entity relevance and sentiment. These targets differ from whole-document source-only review triage.

Closely related decision-model studies also exist. Nandakishore [10] evaluates Laya Typed-Decisions, teacher agreement, calibration and selective escalation on synthetic workflows. Stuani's

engineering benchmark [11] compares Laya, lexical classifiers, encoders and language models on financial complaint routing and fraud flags. Borchers [12] compares adapted decision models with frozen representations and learning curves on research-paper categories. Our task and corpus differ; we do not claim to introduce financial filtering, decision-model comparison or calibration analysis.

Laya's English root checkpoint uses ModernBERT-large and a learned typed-decision head [2, 3]. FinBERT's adaptation to financial text and sentiment [1] makes it a relevant industry comparator, but its sentiment head does not supply review labels. We use its encoder with a task classifier. LoRA [4] and temperature scaling [5] are established methods. Automated annotation still requires task-specific validation [6]; success on another annotation task does not validate this rubric.

3 Task and data

The input is an extracted English document body. Ordinary administrative content is routine; a potentially significant corporate, financial, economic or policy development warrants review. Scheduled publication and sentiment alone do not determine the label. Source identifiers and issuer history are not additional features; metadata support acquisition, grouping and evaluation.

The corpus comprises SEC filings and release exhibits, Federal Reserve releases, ECB institutional/statistical releases, GOV.UK announcements and Wikinews. Training uses 25,000 documents dated through March 2026; selection and calibration use 1,000 and 500 documents from April–June; test uses 2,000 from July–September. Related-document groups cannot cross cohorts. Table 1 shows the substantial SEC dominance. The small news sample does not represent a general financial-news inbox.

We retain earlier definite labels and use Luna Flex with a pinned binary rubric to decide previously unclear references and label new documents. Unclear labels are therefore not mechanically merged into either class. A blinded, balanced 200-document training-only audit checks consistency of retained references against new binary decisions without overriding them. This hybrid protocol measures teacher consistency, not agreement with financial analysts.

Evaluation additions were chosen without labels or model scores while retaining previous memberships. The revised test keeps 1,800 earlier bodies and adds 200. Grouping excludes exact matches, known event aliases and verified near-duplicates; approximate retrieval and inherited groups can miss relationships. Acquisition and extraction details are retained in the public evidence package.

Table 1: Frozen cohort support. The test retains 1,800 earlier bodies; reference labels are revised.

Cohort	Documents	Routine	Review	Groups	SEC	ECB
Train	25000	7173	17827	24696	20740	1175
Selection	1000	283	717	993	889	29
Calibration	500	135	365	493	461	12
Test	2000	457	1543	1967	1870	40

4 Systems and evaluation

4.1 Fixed comparisons and adaptation

Training subsets of 4,000, 16,000 and 25,000 documents are nested, with stable round-robin family/class sampling. All trainable core systems use the same 4,000 bodies; the majority reference counts the same labels without optimization. Native Laya requires no task training. TF-IDF, frozen FinBERT/logistic and both LoRA systems additionally use 16,000 and 25,000 training documents. Every learning-curve point also uses the same 1,500 development labels; training count is not total annotation cost.

TF-IDF learns its vocabulary and transforms from training alone. Logistic regularization and class weighting use the same bounded grid at each training size, selected by selection macro F1 (Appendix A). Frozen FinBERT windows are averaged and L2-normalized before classification. Caches bind body, recipe and source identities; command time includes encoding and classifier search.

We use pinned `convaiinnovations/laya` English root and `ProsusAI/finbert` checkpoints; full revisions accompany the evidence. Adapted neural systems use seed 42, unweighted document-level cross-entropy and up to three epochs. Raw selection macro F1 chooses the checkpoint; epoch zero is eligible, and ties retain the earlier epoch. Laya head adaptation updates the decision head with a frozen encoder at learning rate 3×10^{-5}. LoRA updates rank-eight encoder adapters and the task head at 10^{-4}, with scaling 16 and adapter dropout 0.05. Matched pre-update states are preserved as supporting controls, avoiding a direct attribution of every SDK-to-adapted change to weight updates.

Laya adaptation uses a 1,024-token context with overlapping windows and averages their logits. FinBERT uses a 512-token context with non-overlapping source windows and averages their embeddings before classification. Both retain all normalized source tokens and use one document-level loss; no window labels are inferred. The native Laya SDK uses 512-token windows and limits GPU passes to eight windows, retaining its most-confident long-document aggregation. It therefore differs from adapted Laya in context and pooling as well as fitting. Architectures, parameter counts, pretraining objectives and classifier recipes differ across systems; those comparisons cannot isolate finance pretraining or LoRA as a sole cause.

4.2 Agreement, misses and review workload

Macro F1 averages the routine and review-worthy class F1 values. Exact binary ties route to review. We also report review precision/recall, the 2×2 confusion matrix, and false negatives divided by review-worthy reference support. At the default decision, a document predicted routine is skipped. The skipped fraction quantifies reading removed; its complement is review workload. A low miss rate can reflect more review rather than better discrimination, so both quantities accompany every operational comparison.

Scalar temperature is fit by calibration negative log likelihood. It changes confidence while preserving the default class decision. Binary Brier score is the sum over both class columns, equal to twice the positive-class squared error. We report raw/calibrated scoring arithmetic in the evidence package.

We fit five routine-only skip policies on calibration: empirical review-miss targets of 1% and 5%, and maximum skip budgets of 10%, 20% and 30%. Each chooses the largest feasible calibration skip count without splitting equal-probability ties. Predicted review documents are never skipped by these rules. We report achieved test skip counts and missed references, rather than assume workloads match or that fitting targets are population guarantees.

The fixed primary comparison is 4k Laya LoRA versus 4k word TF-IDF/logistic. We use 2,000 paired whole-group bootstrap draws for percentile 95% intervals and their differences. These intervals condition on a single fitted run; they do not measure training instability or correct supplementary comparisons for multiplicity. Source/length slices are descriptive, with support reported. The two 25k LoRA fits were part of the planned comparisons but completed after the initial report because of training time. Their recipes were fixed in advance and were not adjusted in response to earlier results. All training sizes use the same frozen test cohort.

4.3 Measured resources

Serial GPU jobs use an NVIDIA RTX 3060 (12 GiB); CPU work uses an Intel Core i5-12400F. Warm inference includes document processing but excludes loading, using the same 128 selection bodies. We report mean/P95 latency, throughput, allocated GPU memory and fitting/command durations. Appendix A accounts for preparation and interrupted attempts. Timings describe this machine.

5 Results

5.1 Equal-data system comparison

Laya LoRA has the highest 4k macro F1 point estimate in Table 2. The prespecified Laya-minus-TF-IDF contrast is 0.0199 with paired 95% interval $[-0.0022, 0.0423]$. This interval establishes neither an advantage nor equivalence. At default decisions, Laya skips 156 additional documents but misses 80 more review-worthy references. Its review precision is 94.3% versus 89.9%, while recall is 82.6% versus 87.8%. Higher macro F1 does not imply fewer missed review-worthy references. Laya's warm mean inference latency is 64.9 times the lexical classifier's.

Table 2: Core systems on the revised 2,000-document evaluation. Default decisions; false negatives (FN) are divided by review-reference support. All trainable systems use the same 4k bodies; majority counts those labels, and SDK has no task fit. Warm inference includes full document processing.

System	F1 (95% CI)	FN	Skipped	Mean ms	P95 ms
Training majority	0.436 [0.429, 0.441]	0/1543	0.0%	2.97	3.45
Word TF-IDF/logistic	0.764 [0.743, 0.786]	188/1543	24.6%	4.58	8.16
Frozen FinBERT/logistic	0.718 [0.696, 0.740]	302/1543	30.9%	35.81	113.33
FinBERT LoRA	0.743 [0.721, 0.763]	352/1543	36.5%	35.17	116.72
Laya native SDK	0.542 [0.519, 0.567]	96/1543	8.3%	334.22	1332.02
Laya head	0.609 [0.587, 0.632]	473/1543	37.3%	276.66	1131.21
Laya LoRA	0.784 [0.764, 0.805]	268/1543	32.4%	297.19	1207.55

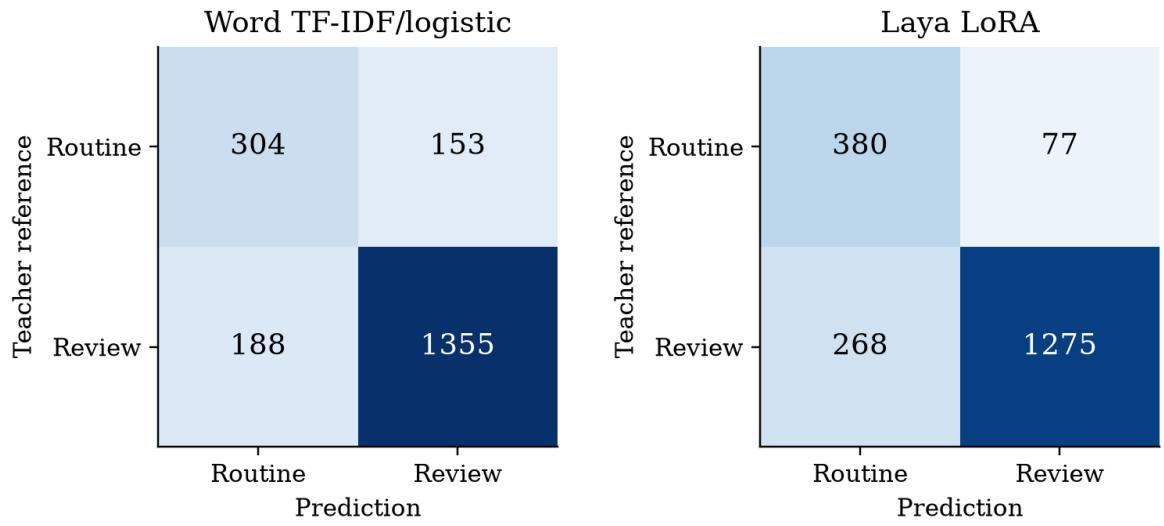


Figure 1: Default two-class confusion matrices for the fixed primary pair. Rows are teacher references; columns are model predictions. Counts sum to 2,000 in each matrix.

5.2 Review workload and missed references

Table 3 applies calibration-selected thresholds unchanged to test. The 1% fitting target and 20% skip budget produce different achieved workloads across systems; neither is a test guarantee. For the primary pair, the 1% calibration miss target reduces skipped test reading to 2.5% for Laya and 6.2% for TF-IDF. Stringent miss control substantially reduces skipping; achieved misses must be read alongside the remaining workload.

Table 3: Test outcomes for the empirical 1% calibration miss target and 20% calibration skip budget. Policies skip only routine predictions, preserving ties; achieved test workloads differ. Rows span the 4k core comparison and the planned 25k fits.

System	1% miss: FN	Skipped	20% skip: FN	Skipped
Training majority	0/1543	0.0%	0/1543	0.0%
Word TF-IDF/logistic	20/1543	6.2%	95/1543	15.8%
Frozen FinBERT/logistic	27/1543	4.0%	159/1543	18.7%
FinBERT LoRA	31/1543	5.3%	144/1543	19.8%
Laya native SDK	14/1543	0.9%	95/1543	8.2%
Laya head	14/1543	1.4%	240/1543	19.6%
Laya LoRA	7/1543	2.5%	124/1543	19.7%
FinBERT LoRA (25k)	9/1543	6.2%	65/1543	17.3%
Laya LoRA (25k)	9/1543	7.1%	62/1543	17.5%

5.3 Adaptation and sample efficiency

Figure 2 shows the nested training-size curves alongside the shared-4k comparison. Source/class composition and optimization steps also vary with size, so the curves do not isolate sample count causally. Intervals condition on fixed fitted runs and are uncorrected for multiple comparisons.

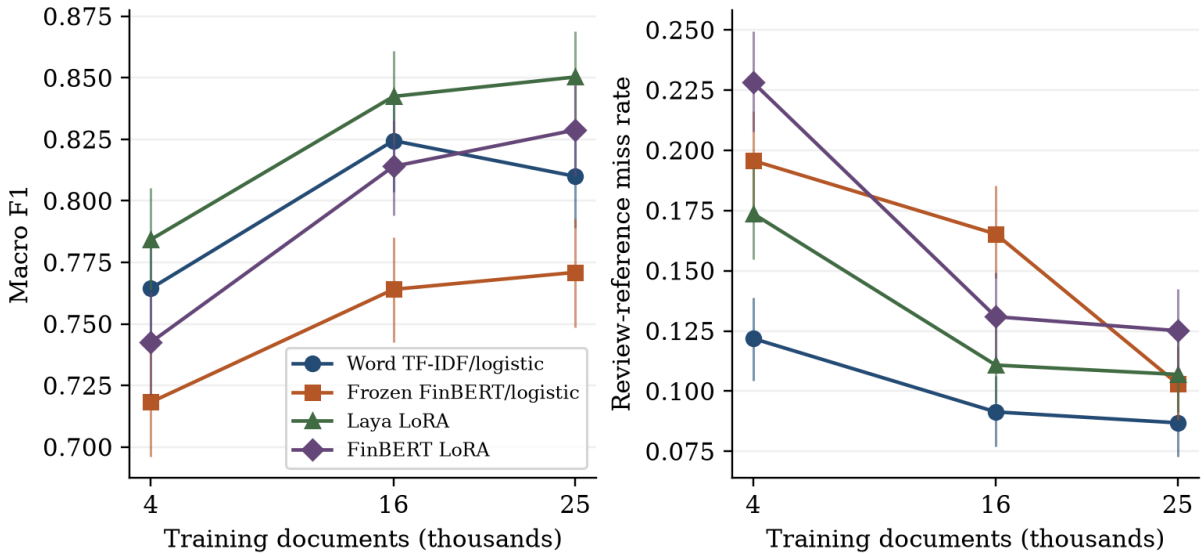


Figure 2: Nested training-size curves. Left: macro F1. Right: review-reference miss rate. Vertical segments are marginal whole-group 95% intervals.

From 16k to 25k, Laya LoRA F1 changes from 0.8424 to 0.8503, misses from 171 to 165, and skipped reading from 28.0% to 28.0%. FinBERT LoRA F1 changes from 0.8140 to 0.8288, with 202 to 193 misses and 28.8% to 29.0% skipped. Command times increase from 20.66 to 37.17 hours for Laya and from 2.51 to 4.36 hours for FinBERT. Tables 3 and 4 report their policy outcomes and paired F1 increments. On the shared selection set, Laya 25k scores 0.8269 versus 0.8112 at epoch one and 0.8540 versus 0.8381 at epoch two; epoch-three scores are 0.86159 and 0.86164. Their final class decisions differ on 56 of 1,000 documents.

Matched pre-update comparisons (Table 5) compare each recipe with its own initial state, separating fitting from context, pooling and initial-head differences. They do not isolate pretraining or equalize classifier budgets.

Table 4: Paired training-size F1 increments for fixed system families. Conditional 95% intervals; comparisons are uncorrected for multiplicity. All sizes use the same frozen test cohort.

System	Training change	Δ F1	95% interval
Word TF-IDF/logistic	4k to 16k	0.0600	[0.0421, 0.0795]
Word TF-IDF/logistic	16k to 25k	-0.0145	[−0.0295, 0.0013]
Frozen FinBERT/logistic	4k to 16k	0.0458	[0.0292, 0.0624]
Frozen FinBERT/logistic	16k to 25k	0.0069	[−0.0098, 0.0230]
Laya LoRA	4k to 16k	0.0581	[0.0428, 0.0735]
FinBERT LoRA	4k to 16k	0.0715	[0.0535, 0.0899]
FinBERT LoRA	16k to 25k	0.0148	[0.0019, 0.0284]
Laya LoRA	16k to 25k	0.0079	[−0.0039, 0.0204]

6 Discussion and limitations

TF-IDF/logistic remains a competitive low-cost baseline. Operational model selection requires an explicit miss/workload objective: macro F1 alone does not determine a suitable routing policy. The training-only reference audit agrees on 93.0% of 200 balanced retained labels. That consistency result cannot substitute for expert reference validity or estimate a population annotation error rate.

Teacher agreement is the main limitation. Training and evaluation share a teacher, and retained and newly decided labels follow different protocols. Expert adjudication is needed to relate classification misses to economically material information. The study supplies neither company/portfolio context nor evidence about trading profitability.

All training-size comparisons share the same test cohort; their paired intervals do not make them independent replications or provide a fresh sample for external generalization. Source imbalance, changing web content, possible pretraining exposure and missed duplicates further limit transfer to industry inboxes.

Fixed recipes and one seed leave optimization variability and the best achievable performance within each model family unresolved. Supplementary intervals are uncorrected for multiple comparisons, and routing thresholds may shift with source mix. Frontier generative classifiers, full-model adaptation and RLCD training are outside this comparison.

7 Conclusion

LoRA improves Laya's teacher agreement relative to matched initial states, with larger gains from 4k to 16k than from 16k to 25k. The latter increment's interval includes zero despite substantially longer fitting. The primary 4k comparison does not establish an F1 advantage over TF-IDF/logistic: Laya skips more documents but misses more review references and costs more to run. Expert labels and fresh representative evaluation data are needed before operational claims.

Code and reproducibility

The public package supplies predictions, references, fitting records, resource measurements and source hashes. Exact model reproduction additionally requires local text snapshots and weights; live source URLs may return changed text. Acquisition records specify attribution and transfer conditions. The study package and 25k LoRA package supply `verify.py` scripts that replay metrics, policies, timing summaries and paired intervals on CPU with Python and NumPy, without bodies, weights or API calls. The original freeze identity is `e6afdaf08581`; full hashes and source identities accompany the packages.

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A Supporting comparisons and provenance

These tables report matched controls and fitting records for the Laya and FinBERT systems. The 25k fits use the same recipes on larger nested subsets; full measurements and per-document arithmetic accompany the public packages. The linear grid uses $C \in \{0.001, 0.01, 0.1, 1, 10, 30, 100\}$ and either no class weighting or balanced weights.

Table 5: Matched pre-update comparisons. The initial adapted pooling/head state is the control; it is distinct from native SDK inference. Larger fixed recipes reuse a matched setup's pre-update control. Intervals are supplementary.

System	Train	Zero F1	Fit F1	Δ F1 interval
Laya head	4k	0.557	0.609	[0.023, 0.083]
Laya LoRA	4k	0.557	0.784	[0.196, 0.259]
FinBERT LoRA	4k	0.187	0.743	[0.535, 0.574]
FinBERT LoRA	16k	0.187	0.814	[0.607, 0.646]
Laya LoRA	16k	0.557	0.842	[0.255, 0.316]

Table 6: Completed neural configurations. All use seed 42 and at most three epochs; epoch zero is eligible. A context limit includes task/model overhead, rather than denote full source coverage. All normalized source tokens are windowed.

System	Train	Context	Trainable	Selected epoch
Laya head	4k	1024	26248193	2
Laya LoRA	4k	1024	27165697	3
FinBERT LoRA	4k	512	297986	3
FinBERT LoRA	16k	512	297986	3
Laya LoRA	16k	1024	27165697	3

Table 7: Recorded resources. Command time includes preparation, fitting and development evaluation. Fit phase is the neural history or linear preparation/search duration, including selection overhead. Fit GiB records neural-fit peaks only; '–' denotes an unavailable measurement. Inference peaks can include resident fitting state. Frozen feature/cache costs are included in command time and itemized in the evidence.

System	Train	Command s	Fit phase s	Fit GiB	Infer GiB
Training majority	Count 4k	6.3	0.2	–	0.00
Word TF-IDF/logistic	4k	22.9	15.9	–	0.00
Frozen FinBERT/logistic	4k	219.1	199.1	–	0.51
Laya native SDK	None	812.9	–	–	2.35
Laya head	4k	9225.8	8541.7	2.37	2.10
Laya LoRA	4k	17014.8	16278.2	3.24	2.12
FinBERT LoRA	4k	2209.4	2099.5	0.78	0.52
Word TF-IDF/logistic	16k	59.5	51.8	–	0.00
Word TF-IDF/logistic	25k	97.2	89.8	–	0.00
Frozen FinBERT/logistic	16k	485.9	465.4	–	0.51
Frozen FinBERT/logistic	25k	481.5	459.2	–	0.51
FinBERT LoRA	16k	9043.6	8951.9	0.79	0.52
Laya LoRA	16k	74374.3	73637.6	3.24	2.12

Laya native SDK also used 886.0 seconds in prior interrupted attempts. These attempts are excluded from the successful-command column; cumulative elapsed time and attempt receipts remain in the evidence package.